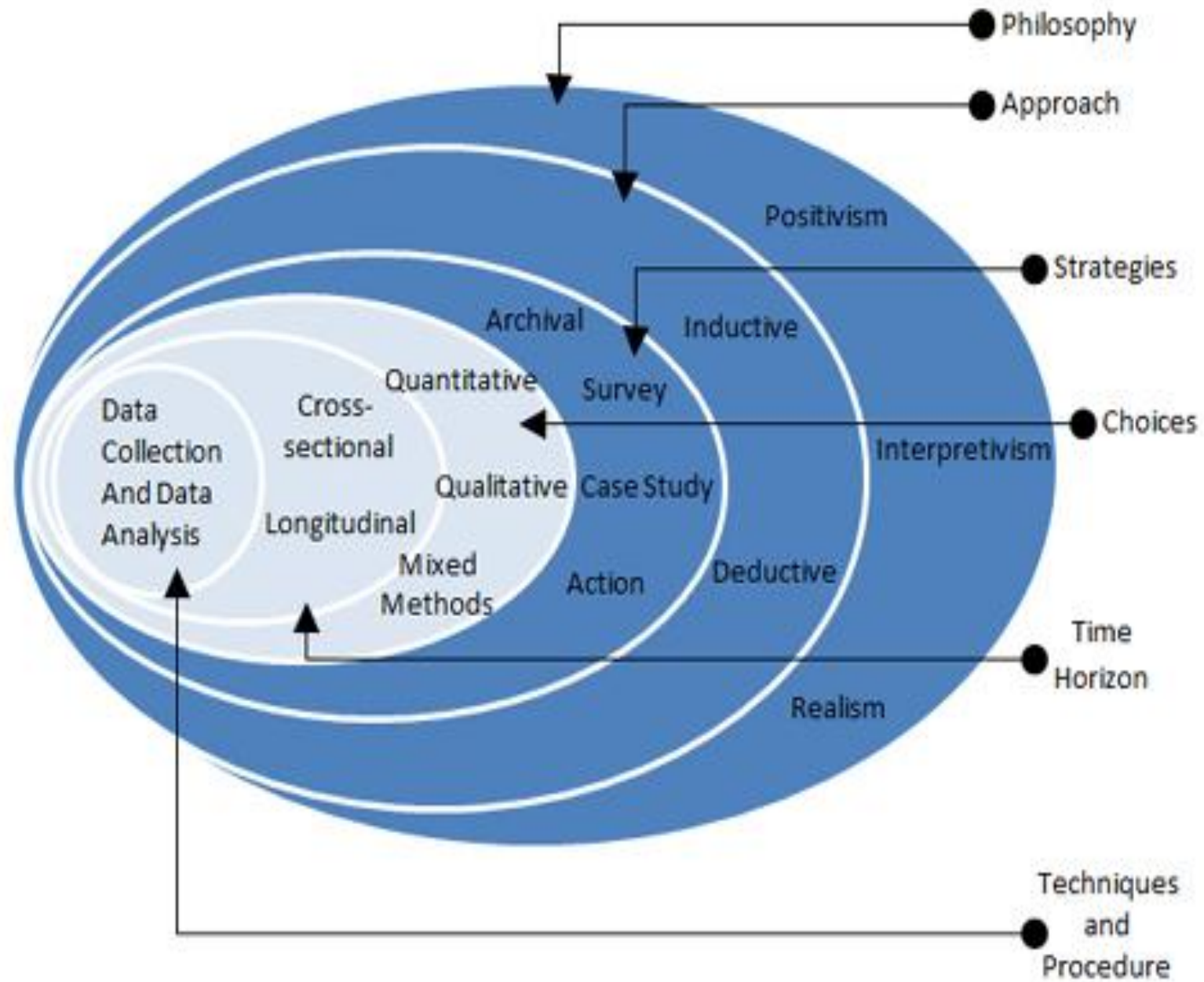


RESEARCH METHODOLOGY

Research Methodology: is a systematic way to solve a problem. It defines how a research is carried out and by which a knowledge is gained.

The Research Onion Model (Source: Saunders et al. 2007)



Layer 1: Philosophical Stances

In order to understand the philosophical stances, there are three main stances;

1- Positivism: This position is based on the idea that scientific knowledge is the true or acceptable knowledge of the world and is characterized by the testing of the hypotheses (or research questions) derived from the existing theory of knowledge.

2- Realism: The philosophy of realism questions reliability of the scientific knowledge and maintains that all theories can be revised and more reliable results can be obtained through continual research and application of new methods of research. Hence, new research methods contribute to the acceptable knowledge in this case.

3- Interpretivism: This philosophy emphasizes the use of qualitative analysis over quantitative or statistical analysis to obtain the results. The interpretivist researcher is often seen playing an important role in making sense of and interpreting the collected data. Therefore, it would not be wrong to say that interpretivism incorporates human interest into a research study and recognizes differences between people.

Layer 2: Research Approach

A research approach can be **Deductive** or **Inductive**;

1- Deductive: This research approach tends to flow from generic to specific. Any researcher using deductive reasoning would start with theory and move on to research question or hypothesis which is tested through data collection afterwards. In the end, findings derived from the collected data would either confirm or reject the research question or hypothesis.

2- Inductive: This type of reasoning or approach is often used or helpful when there is little research available on a topic. Unlike deductive approach, a researcher has to move in the opposite direction from research question to observation and description to analysis before finally getting on to theory.

Layer 3: Research Strategies

The third layer of the research onion is concerned with how a researcher plans to collect data. Following are some research strategies;

- Experiment
- Survey
- Case Study
- Action Research
- Grounded Theory
- Ethnography
- Archival Research

Experiment – Scientific and rigid in structure, experimental designs are meant to test causal effects of phenomena on a group. The data collected using this method could be statistically analyzed.

Survey – A survey design often results from a deductive approach and is capable of collecting large volume of data that is apt for statistical analysis. The collected data helps answer the research question or hypothesis in a cost-effective manner.

Case study – This type of research design provides unique examples of real people or cases in real situations. The number of such case studies is often restricted for drawing clear conclusions from the data.

Layer 4: Research Choices

Three methods to choose from when working with this specific layer of the research onion. These are listed below:

Quantitative – This type of method takes place when the researcher goes about collecting quantitative data based on the decisions made in the previous stage.

Qualitative – This type of method takes place when the researcher goes about collecting qualitative data based on the decisions made in the previous stage.

Mixed-method – This means that both qualitative and quantitative data collection techniques are used.

Layer 5: Time Horizons

In this layer, researcher has to decide about the timeframe of the research. There are three options for a researchers to choose from;

1- **Cross-sectional:** In this timeframe, the data is collected from different cross-sections (individual/entity) at a fixed time.

2- **Longitudinal:** In this timeframe, the data is collected from a fixed cross-sections (individual/entity) at a different time.

3- **Panel:** In this timeframe, the data is collected from different cross-sections (individual/entity) at a different time.

Layer 6: Techniques & Procedures

The final layer of the research onion looks at data collection and analysis.

Basic Terms and Concepts

Population: the universe of units from which the sample is to be selected.

Sample: the segment of population that is selected for investigation

Sampling frame: list of all units

Representative sample: a sample that reflects the population accurately

Sample bias: distortion in the representativeness of the sample

Sampling Techniques

Probability Sampling: this Sampling technique uses randomization to make sure that every element of the population gets an equal chance to be part of the selected sample. It's alternatively known as random sampling.

Non-Probability Sampling: it does not rely on randomization. This technique is more reliant on the researcher's ability to select elements for a sample. This type of sampling is also known as non-random sampling.

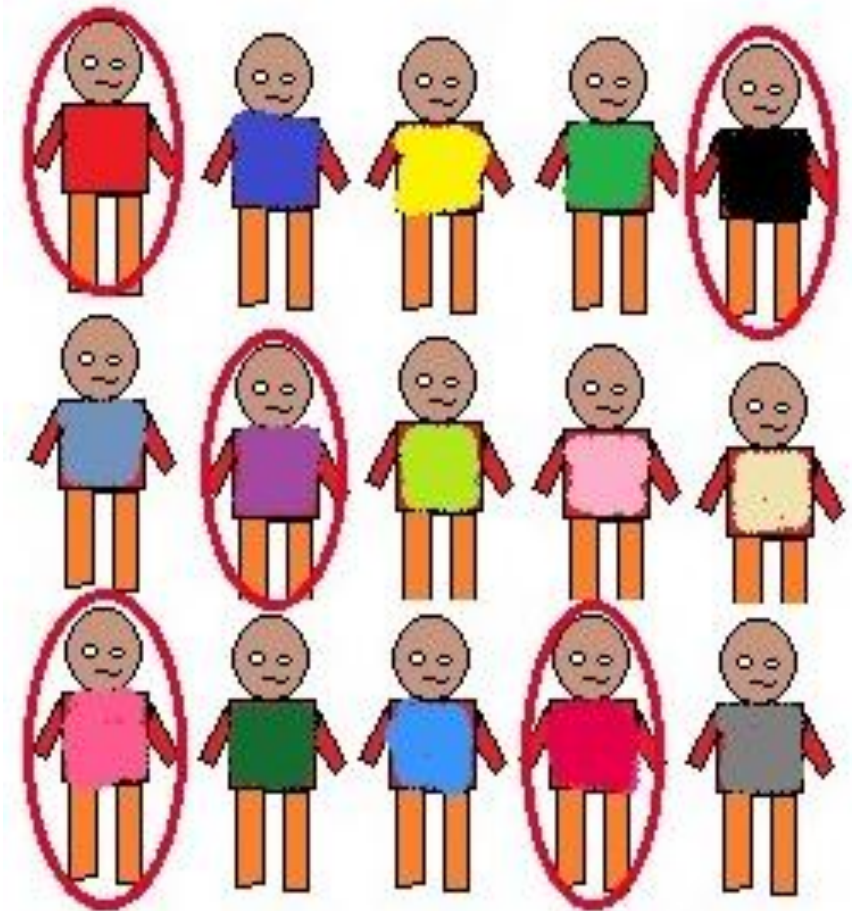
Types of Probability Sampling

- Simple random sample
- Systematic sample
- Stratified random sample
- Cluster sample

Simple Random Sampling

Every element has an equal chance of getting selected to be the part sample. It is used when we don't have any kind of prior information about the target population.

For example: Random selection of 20 students from class of 50 student. Each student has equal chance of getting selected. Here probability of selection is $1/50$.



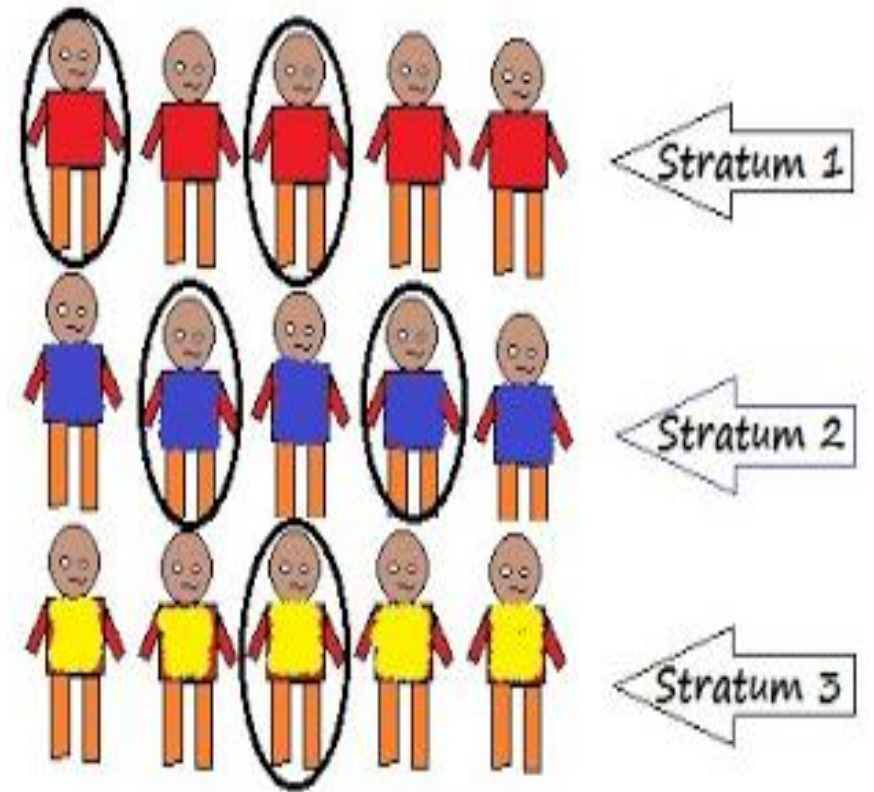
Systematic Random Sampling

Here the selection of elements is systematic and not random except the first element. Elements of a sample are chosen at regular intervals of population. All the elements are put together in a sequence first where each element has the equal chance of being selected.



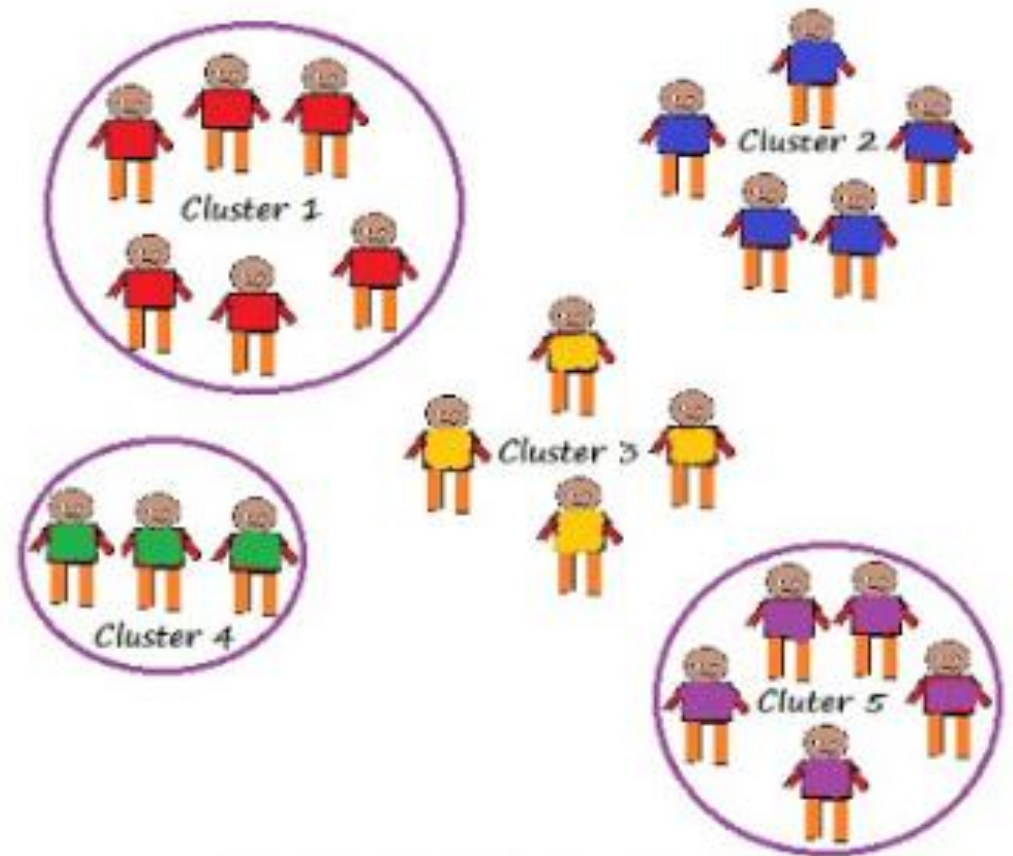
Stratified Random Sampling

This technique divides the elements of the population into small subgroups (strata) based on the similarity in such a way that the elements within the group are homogeneous and heterogeneous among the other subgroups formed. And then the elements are randomly selected from each of these strata. We need to have prior information about the population to create subgroups.



Cluster Random Sampling

Our entire population is divided into clusters or sections and then the clusters are randomly selected. All the elements of the cluster are used for sampling. Clusters are identified using details such as age, gender, location etc.



Types of Non-Probability Sampling

- Convenience Sampling
- Purposive Sampling
- Snowball Sampling
- Quota Samplings

Convenience Sampling

Here the samples are selected based on the availability. This method is used when the availability of sample is rare and also costly. So based on the convenience samples are selected.

For example: Researchers prefer this during the initial stages of survey research, as it's quick and easy to deliver results.

Purposive Sampling

This is based on the intention or the purpose of study. Only those elements will be selected from the population which suits the best for the purpose of our study.

For Example: If we want to understand the thought process of the people who are interested in pursuing master's degree then the selection criteria would be "Are you interested for Masters in..?"

Snowball Sampling

This technique is used in the situations where the population is completely unknown and rare. Therefore we will take the help from the first element which we select for the population and ask him to recommend other elements who will fit the description of the sample needed.

Quota Sampling

This type of sampling depends on some pre-set standard. It selects the representative sample from the population.

For example: If our population has 45% females and 55% males then our sample should reflect the same percentage of males and females.